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#include <stdio.h>

#include <stdlib.h> // For exit()


#define MAX_SIZE 5 // Define the maximum size of the queue


int cqueue_array[MAX_SIZE];

int front = -1;

int rear = -1;


// Function to check if the queue is full
int isFull() {

    if ((front == 0 && rear == MAX_SIZE - 1) || (front == (rear + 1) % MAX_SIZE)) {

        return 1;

    }

    return 0;

}


// Function to check if the queue is empty
int isEmpty() {

    if (front == -1) {

        return 1;

    }

    return 0;

}
```

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// Function to enqueue (insert) an element
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void enqueue(int data) {  
    if (isFull()) {  
        printf("Queue overflow: Cannot insert %d, queue is full.\n", data);  
        return;  
    }  
    if (front == -1) { // If the queue was empty  
        front = 0;  
    }  
    rear = (rear + 1) % MAX_SIZE;  
    cqueue_array[rear] = data;  
    printf("Element %d inserted.\n", data);  
}
```

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// Function to dequeue (remove) an element
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int dequeue() {  
    int data;  
    if (isEmpty()) {  
        printf("Queue underflow: Cannot dequeue, queue is empty.\n");  
        return -1; // Indicate an error or invalid operation  
    }  
    data = cqueue_array[front];  
    if (front == rear) { // If there was only one element  
        front = -1;  
        rear = -1;  
    }  
}
```

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    } else {

        front = (front + 1) % MAX_SIZE;

    }

    printf("Dequeued element: %d\n", data);

    return data;

}

// Function to display the queue elements

void display() {

    if (isEmpty()) {

        printf("Queue is empty.\n");

        return;

    }

    printf("Queue elements: ");

    int i = front;

    while (i != rear) {

        printf("%d ", cqueue_array[i]);

        i = (i + 1) % MAX_SIZE;

    }

    printf("%d\n", cqueue_array[rear]); // Print the last element

}

int main() {

    int choice, value;

```

```
while (1) {

    printf("\nCircular Queue Operations:\n");

    printf("1. Enqueue\n");

    printf("2. Dequeue\n");

    printf("3. Display\n");

    printf("4. Exit\n");

    printf("Enter your choice: ");

    scanf("%d", &choice);


    switch (choice) {

        case 1:

            printf("Enter the value to enqueue: ");

            scanf("%d", &value);

            enqueue(value);

            break;

        case 2:

            dequeue();

            break;

        case 3:

            display();

            break;

        case 4:

            exit(0);

        default:

            printf("Invalid choice. Please try again.\n");
```

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    }  
}  
return 0;  
}
```